

AURA V4 — What Is Wrong, Red Flags, What We Fix, and What We Add for AURA V5

After reviewing the notebook + Csaba's expert feedback + Pieter's strategic direction, the conclusion is very clear:

AURA V4 is strong as a prototype

but

weak as a commercial valuation engine

This is not failure.

This is exactly the normal transition from:

academic prototype → commercial infrastructure

and this is precisely what Csaba was testing.

PART 1 — THE BIG RED FLAGS (Critical Problems)

RED FLAG 1 — Fair Price Is Not Fully Defensible

Problem

Current target = mostly listing price behavior

but:

```
asking price ≠ transaction price  
listing price ≠ market value
```

This is the biggest problem.

If fair_price is learned from inflated listing prices,
then AURA is predicting:

“market bias”

instead of:

“real fair value”

This is exactly the chicken-and-egg problem Csaba raised.

RED FLAG 2 — Leakage Suspicion

Problem

Top feature:

```
city_prestige = +0.29056
```

Second feature:

```
price_vs_local_sqm = +0.09566
```

These are dangerous.

Why?

Because they may contain target information disguised as features.

Especially:

price_vs_local_sqm

because it uses asking price itself.

That is extremely dangerous.

It can create:

artificial R^2 inflation

and fake confidence.

This is likely the strongest technical red flag.

RED FLAG 3 — Geography Is Too Coarse

Problem

Model mostly understands:

- district
- city
- market tier

But:

Lisbon is not one market

District-level intelligence is too weak.

A luxury street and a bad street inside Lisbon are not the same.

This destroys true valuation precision.

Csaba directly flagged this.

RED FLAG 4 — Validation Is Statistical, Not Real

Problem

Current validation:

- R^2
- MAE
- RMSE
- MAPE
- OOF

Missing:

exogenous validation

Questions missing:

- Did it actually sell?
- Time on market?
- Closing transaction price?
- Seller recommendation success?
- Was WAIT better than SELL_NOW?

Without this:

it is still academic.

Not commercial.

Huge issue.

RED FLAG 5 — Confidence Score Is Misleading

Problem

Current confidence uses:

- completeness
- freshness
- source reliability

but not true:

model uncertainty

Example:

```
confidence = 0.975
```

This sounds like:

"I am 97.5% sure"

but actually means:

"data completeness is high"

This is dangerous for trust.

RED FLAG 6 — Overfit Gap Exists

Problem

```
00F R2 = 0.8384  
VAL R2 = 0.7772  
Gap = +0.0612
```

This is moderate overfitting.

Not catastrophic

but enough to justify Csaba's suspicion.

Especially combined with leakage risk.

RED FLAG 7 — Sparse Critical Layers

Problem

Very weak data coverage:

- Calibration → 18.4%
- HPI → 21.9%
- Macro → 21.9%
- Regulatory → 93% missing

This means many “intelligence layers” are conceptually strong but practically weak.

AURA looks stronger than the real data support.

This must be fixed.

RED FLAG 8 — SELL_NOW Almost Never Happens

Problem

```
SELL_NOW = 2.0%  
WAIT = 53.6%  
REDUCE_PRICE = 41.1%
```

This means:

Decision Engine may be too conservative or poorly calibrated.

Commercially:

users want action.

Not permanent waiting.

This needs recalibration.

RED FLAG 9 — Scenario Engine Is Broken

Problem

```
Best:  
Conservative = 123,451  
Aggressive = 3
```

This is clearly unhealthy.

It means scenario logic is too biased.

This destroys strategic usefulness.

Must be rebuilt.

RED FLAG 10 — Explainability Is Still “Pretty ML”

Problem

Users still hear:

```
Because model says so
```

instead of:

```
Because:  
district trend ↑  
macro pressure ↓  
buyer demand strong  
regulatory pressure high
```

Explainability must become:

decision intelligence

not model storytelling.

PART 2 — WHAT WE FIX IMMEDIATELY

FIX 1 — Rebuild Fair Price Logic

Priority = Highest

Move from:

listing-price learning

to

transaction-price truth

using:

- confidential sold-price data
- Helios
- transaction registries
- not listing portals only

This is non-negotiable.

FIX 2 — Full Leakage Audit

Audit:

- city_prestige
- district_prestige

- price_vs_local_sqm
- seller_advantage
- target leakage
- OOF generation validity
- split strategy

Some features may need complete removal.

FIX 3 — Geography Upgrade

Move to:

- municipality
- parish
- postcode
- street proxies
- walkability
- transit
- schools
- beach distance
- premium zone clustering

This is massive for V5.

FIX 4 — Real-World Validation Layer

Build:

Outcome Validation Engine

Track:

- sold vs unsold

- days on market
- final closing price
- negotiation success
- seller action performance

This is what investors trust.

FIX 5 — Confidence Rebuild

Split into:

Data Confidence

and

Prediction Confidence

Never merge both.

This is critical.

FIX 6 — Decision Engine Recalibration

Rebalance:

- SELL_NOW
- WAIT
- HOLD
- REDUCE_PRICE

using real seller outcomes.

Not static thresholds.

PART 3 — NEW PHASES TO ADD (AURA V5)

NEW PHASE A — Transaction Truth Engine

The most important new phase

Role:

true sold-price intelligence

This becomes:

the heart of AURA

not listing data.

NEW PHASE B — Exogenous Validation Engine

Tracks:

- sell-through success
- pricing success
- recommendation accuracy
- market reaction

This is what proves product value.

NEW PHASE C — Micro-Geography Intelligence Engine

Not district.

Real geo intelligence:

- postcode
- street
- municipality

- neighborhood scoring

Huge priority.

NEW PHASE D — Institutional Explainability Layer

Built for:

- investors
- brokers
- banks
- sellers

with:

decision reasoning

not ML explanation.

NEW PHASE E — Proprietary AI Assistant (Open Source LLM)

Replace Gemini with:

proprietary enterprise AI assistant

Likely:

- Qwen
- Mistral
- Llama-class

with:

- internal knowledge base

- governance
- audit trail
- explainability
- no API dependency

This is Week 12 strategic priority.

NEW PHASE F — Seller Feedback Loop

Capture:

- broker feedback
- seller outcomes
- rejected recommendations
- human overrides

This creates:

the data flywheel

and the moat.

Massive strategic advantage.

FINAL TRUTH

AURA V4 solved:

"Can we build this?"

AURA V5 must solve:

"Can the market trust this?"

That is the real mission.

And that is exactly what Csaba was pushing you toward.

My Honest Strategic View

The highest priority is NOT:

dashboard

NOT frontend

NOT prettier charts

NOT another model

It is:

proving fair price is real

because if fair price is weak,

everything collapses.

That is where AURA V5 truly begins.